Surya Tej

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Summary

Embedded Software Developer building RTOS firmware for smart home automation on ESP32, ESP8266, and Nuvoton, from Wi-Fi switches and dimmers to a Matter-based IoT Hub. Smart India Hackathon 2023 Finalist as Team Leader for Smart Microgrid as a Service; selected for Grand Contest 2024 with a LiDAR-enabled smartwatch for the visually impaired.

Technical Skills

Languages & Tools: C, C++, Embedded C, Python, HTML, CSS; Arduino IDE, PlatformIO, CMake, Git/GitHub.
Embedded & IoT: ESP32, ESP8266, Nuvoton, RTOS, Matter, Wi-Fi, UART, I2C, SPI, GPIO, Hardware-Software Integration.
Engineering: System Design, Debugging, Prototyping, Hardware & System Testing, Performance Optimization, Network Analysis.

Experience

PAT IoT Solutions — Hyderabad, India
Embedded Software Developer — Sep 2025 – Present

- Develop end-to-end firmware for hub, master, and slave devices, including DALI and other lighting controllers.
- Implement RTOS-based architectures and communication protocols, and debug hardware-software integration on real devices.
- Collaborate with hardware, firmware, and product teams to build reliable connected automation for production.
- Beyond embedded work, designed and deployed the PAT IoT Solutions website using Claude AI.

CSR Electronologies — Hyderabad, India
Firmware Engineer Intern — May 2025 – Sep 2025

- Developed firmware for Wi-Fi-enabled smart switches and real-time control systems on ESP32 and ESP8266.
- Integrated peripherals over UART, I2C, SPI, and GPIO; worked on RTOS task management and scheduling.
- Built and tested firmware in Arduino IDE and PlatformIO through prototyping, integration, and validation.

Projects

Smart Home Automation & IoT Hub — ESP32, ESP8266, Nuvoton, RTOS, Matter Protocol — 2025 – Present

- Built firmware for Wi-Fi smart switches and dimming controllers supporting Phase-Cut, PWM, DALI, and Analog.
- Developing a Matter-based IoT Hub that handles 25+ devices and works with Alexa, Siri, and Google Assistant.

Smart Microgrid as a Service — Python, Machine Learning | Smart India Hackathon 2023 Finalist — December 2023

- Team Leader of the only team from KITS Warangal to reach the Smart India Hackathon 2023 finals.
- Built a platform that manages energy generation, storage, and distribution according to demand, with real-time monitoring.
- Added analytics and user-facing suggestions to optimise power usage and efficiency.

IoT Telemetry Device Agent — C++, CMake, Embedded Systems — July 2026

- Simulated firmware agent that keeps sensor data safe on flash and uploads it reliably over an unreliable link within a 256 KB RAM / 8 MB flash budget, with no data loss through outages or power cuts.

FQ-CoDel-Based Scheduling for IoT Traffic in WiFi Networks — NS-3 Simulator — April 2025

- Implemented FQ-CoDel fair-queuing congestion control to improve traffic prioritisation for IoT devices in WiFi networks.
- Simulated in NS-3 and NetAnim, evaluating latency, throughput, and congestion behaviour under different network conditions.

Smart and Sustainable Surveillance — Python, OpenCV, YOLO — March 2024

- Real-time webcam and CCTV object, face, and motion detection with OpenCV and YOLO, plus an event-tracking interface.

Achievements & Activities

- Smart India Hackathon 2023 Finalist and Team Leader, the only team from KITS Warangal to reach the finals.
- Selected for Grand Contest 2024 with a LiDAR-enabled smartwatch for the visually impaired.
- IDE Bootcamp 2024, NIT Goa; Outreach Program on AI & ML in IoT and Its Applications, NIT Warangal; Technical Event Organiser, Sumshodhini'24, KITS Warangal.

Education

Kakatiya Institute of Technology and Science — Warangal, India
B.Tech in Computer Science Engineering (Internet of Things) — CGPA: 7.03/10 — Nov 2021 – 2025